

SQL Commands with Examples

Introduction to SQL

SQL (Structured Query Language) is a standard language for managing and manipulating relational databases. It is used to perform tasks such as updating data on a database, retrieving data from a database, and creating new databases.

Categories of SQL Commands

SQL commands are broadly categorized into the following:

- **DDL (Data Definition Language):** Used to define and modify the structure of a database.
- **DML (Data Manipulation Language):** Used for managing data within schema objects.
- **DQL (Data Query Language):** Used to retrieve data
- **DCL (Data Control Language):** Used to manage permissions and access control.
- **TCL (Transaction Control Language):** Used to manage transactions in a database.

DDL Commands

DDL commands are used to create, modify, and delete database objects like tables, indexes, and users.

Command	Description	Example
CREATE	Creates a new database, table, or other database objects.	<pre>CREATE DATABASE EmployeesDB;CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, FirstName VARCHAR(50), LastName VARCHAR(50), DepartmentID INT);</pre>

Command	Description	Example
ALTER	Modifies an existing database object.	ALTER TABLE EmployeesADD Email VARCHAR(100);
DROP	Deletes an entire database, table, or other database objects.	DROP TABLE Employees;DROP DATABASE EmployeesDB;
TRUNCATE	Removes all records from a table, but not the table structure itself.	TRUNCATE TABLE Employees;
RENAME	Renames an existing database object.	ALTER TABLE EmployeesRENAME TO Staff;

DML Commands

DML commands are used to manipulate data within the database.

Command	Description	Example
INSERT INTO	Adds new rows of data into a table.	INSERT INTO Employees (EmployeeID, FirstName, LastName, DepartmentID, Email)VALUES (1, 'John', 'Doe', 101, 'john.doe@example.com');
UPDATE	Modifies existing data in a table.	UPDATE Employees SET Email = 'jane.smith@example.com' WHERE EmployeeID = 2;
DELETE FROM	Removes rows of data from a table.	DELETE FROM Employees WHERE EmployeeID = 3;

DQL Commands

DQL command is used to retrieve data for the database.

Command	Description	Example
SELECT	Retrieves data from one or more tables.	<pre>SELECT * FROM Employees;SELECT FirstName, LastName FROM Employees WHERE DepartmentID = 1;</pre>

DCL Commands

DCL commands manage user access and permissions to the database.

Command	Description	Example
GRANT	Gives users permission to perform specific operations on database objects.	<pre>GRANT SELECT, INSERT ON Employees TO 'user1'@'localhost';</pre>
REVOKE	Removes previously granted permissions.	<pre>REVOKE DELETE ON Employees FROM 'user1'@'localhost';</pre>

TCL Commands

TCL commands are used to manage transactions, ensuring data integrity.

Command	Description	Example
COMMIT	Saves all changes made during the current transaction.	<pre>UPDATE Accounts SET Balance = Balance - 100 WHERE AccountID = 1;UPDATE Accounts SET Balance = Balance + 100 WHERE AccountID = 2;COMMIT;</pre>
ROLLBACK	Undoes all changes made during the current transaction.	<pre>UPDATE Orders SET Status = 'Cancelled' WHERE OrderID = 5;ROLLBACK;</pre>
SAVEPOINT	Sets a point within a transaction to which you can later roll back.	<pre>SAVEPOINT SP1;-- Some operations ROLLBACK TO SP1;</pre>

Conclusion

Understanding SQL commands is fundamental for anyone working with relational databases. This document provides a basic overview and examples of the most commonly used DDL, DML, DCL, and TCL commands. For more advanced usage and specific database systems, further documentation and practice are recommended.